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https://github.com/surabhi84ard/Mini/tree

Experiment: POS-based LED Blink using VAMAN Board

Course: Digital Logic Design using Verilog HDL

Objective

To design and implement a Verilog module that computes a function based on its **Product of Sums (PoS)** form and uses it to control the blinking of an LED on the VAMAN FPGA board.

Apparatus Required

- VAMAN FPGA Development Board (based on Spartan-6)
- USB JTAG Programmer
- LEDs and Switches on board
- Xilinx ISE / Vivado Software

Logic Description

The Boolean function is given as:

$$f = (b + c)(a + \bar{c})$$

The LED will blink only when the output of f is logic HIGH. If $f = 0$, the LED remains OFF.

Truth Table

A	B	C	Expression	F
0	0	0	$(b+c)(a+\bar{c})$	0
0	0	1	$(0+1)(0+\bar{1}) = 1 \times 0$	0
0	1	0	$(1+0)(0+\bar{0}) = 1 \times 1$	1
0	1	1	$(1+1)(0+\bar{1}) = 1 \times 0$	0
1	0	0	$(0+0)(1+\bar{0}) = 0 \times 1$	0
1	0	1	$(0+1)(1+\bar{1}) = 1 \times 1$	1
1	1	0	$(1+0)(1+\bar{0}) = 1 \times 1$	1
1	1	1	$(1+1)(1+\bar{1}) = 1 \times 0$	0

Simplified Boolean Expression

$$f = (b+c)(a+\bar{c})$$

No further simplification is required since it is already in minimized PoS form.

Verilog Code

```
module pos_blink(
    input wire clk,      // clock input
    input wire a,        // input a (example: connect to switch or pin)
    input wire b,        // input b (example: connect to switch or pin)
    input wire c,        // input c (example: connect to switch or pin)
    output reg led       // LED output
);

reg [23:0] cnt = 0;

// Compute minimized PoS: (b + c) & (a + ~c)
wire f;
assign f = (b | c) & (a | ~c);

// Blink logic: LED blinks only when the PoS output f is high
always @(posedge clk) begin
    if (f) begin
        cnt <= cnt + 1;
        if (cnt == 24'd5000000) begin // adjust for desired blink rate
            led <= ~led;
            cnt <= 0;
        end
    end else begin
```

```
        led <= 0;     // LED off when PoS output is low
        cnt <= 0;
    end
end

endmodule
```

Listing 1: Verilog code for PoS-based LED blink

Pin Configuration (VAMAN Board Example)

Signal	FPGA Pin	Description
clk	P56	On-board 50 MHz Clock
a	SW0	Input Switch 0
b	SW1	Input Switch 1
c	SW2	Input Switch 2
led	LED0	Output LED 0

Conclusion

The circuit successfully implements the given PoS expression. The LED blinks at a fixed rate whenever the PoS output $f = 1$, and remains off otherwise.

1 Figures

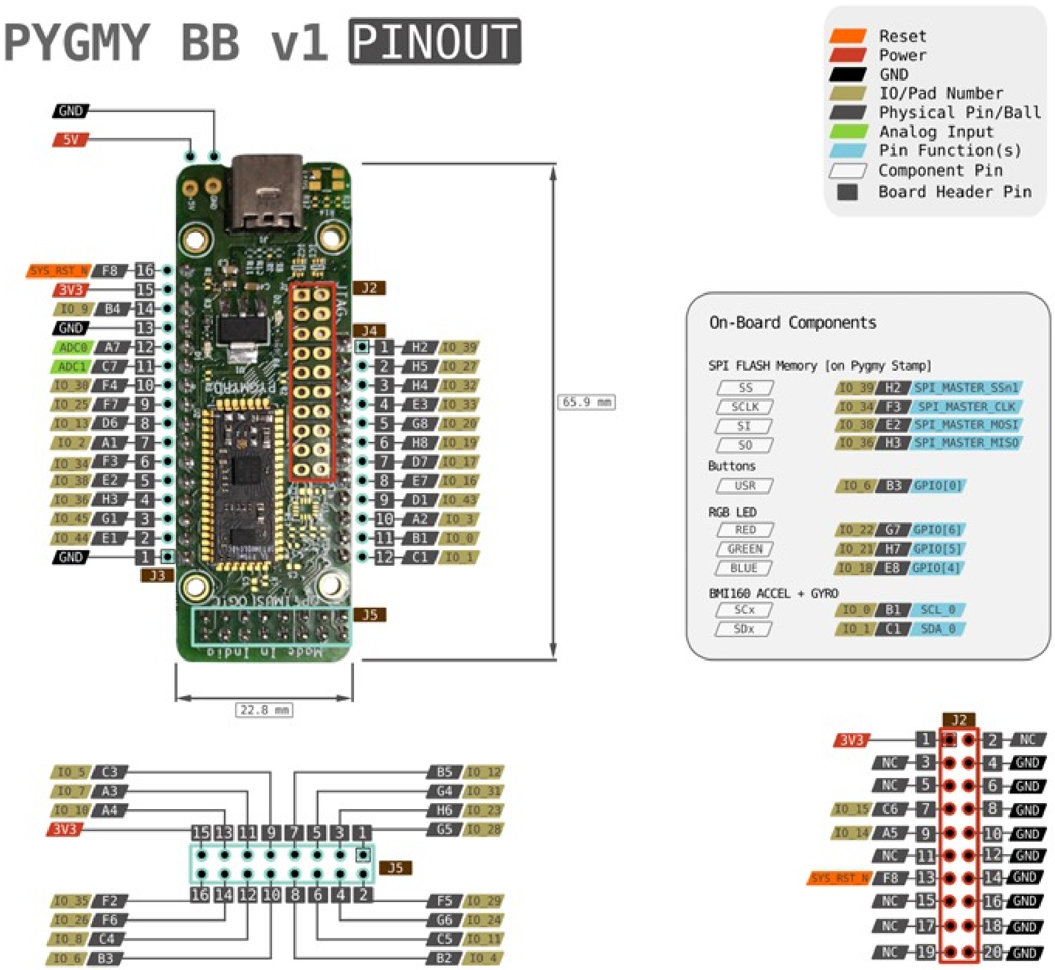

Figure 1: Pygmy BB v1 Pin Diagram